

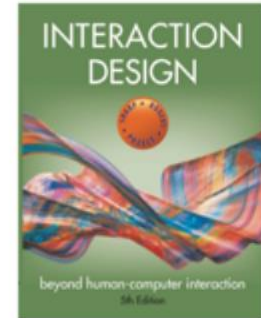


**IN
PARTNERSHIP
WITH
PLYMOUTH
UNIVERSITY**

**PUSL3122 HCI, Computer
graphics and visualisation**

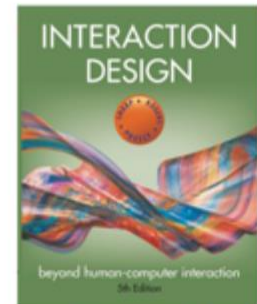
Topics covered last lecture (lecture 1)

- Interaction design, User experience, User experience goals, design principles
- IxD activities: **Discovering** requirements, **Designing** alternatives, **Prototyping and Evaluating**
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapters 1 and 2.



Chapter 1

WHAT IS INTERACTION DESIGN?

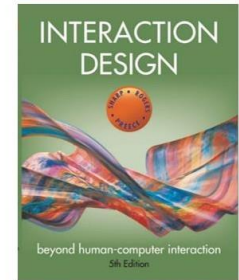


Chapter 2

THE PROCESS OF
INTERACTION DESIGN

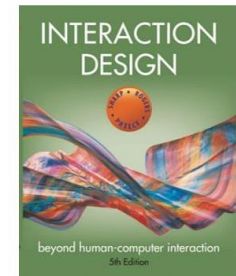
Topics covered this lecture (Lecture 2)

- **What data** needs to be **gathered** 
 - How to **plan** and **run** a **successful** data gathering program.
 - How to **plan** and **run** an interview, design a simple questionnaire and plan and carry out an observation.
- What do I **do with my data**?
 - Difference between **qualitative** and **quantitative** data and analysis; how to present findings
 - Software packages available to help with analysis
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapters 8 and 9.



Chapter 8

DATA GATHERING



Chapter 9

Data Analysis, Interpretation,
and Presentation



What data should I gather ?



Five key issues

- Setting **goals**
 - Decide how to analyze data once collected
- Identifying **participants**
 - Decide from whom to gather data
 - How many participants are needed
- **Relationship** with participants (it's really important!)
 - Clear and professional
 - Informed consent when appropriate
- **Triangulation (keep scope wide)**
 - Look at data from **more than one perspective**
 - Collect **more than one type of data**, for instance, qualitative data from experiments and qualitative data from interviews
- **Pilot** studies
 - Small trial of main study (hence our **MOCK Ux session**)

Data recording



yes



no

Assuming you have ethical approval to carry out the study and consent has been given by the participant

- How will you **record the data** during your session?
 - **Notes**, audio, video, and photographs can be used individually or in combination
 - Notes plus photographs
 - Audio plus photographs
 - Video
 - **Different challenges** and **advantages** with each type of data recording



Interviews

- **Unstructured**: Not directed by a script. **Rich** but not **replicable**.
- **Structured**: Tightly scripted, often like a questionnaire. **Replicable** but may **lack richness**.
- **Semi-structured**: Guided by a script, but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.
 - My recommendation!
- Focus groups: A **group interview**
 - These work well as interactions between the participants generates **discussion** and sometimes **debate** (which usually highlights area of confusion and issues)

Interview questions

- Two types:
 - ‘**Closed questions**’ have a predetermined answer format, for example, ‘yes’ or ‘no’
 - ‘**Open questions**’ do not have a predetermined format
- Closed questions are easier to analyze
- Avoid:
 - **Long** questions (avoid lots of text)
 - Compound sentences — **split** them into two
 - **Jargon** and language that the interviewee may not understand (put participant at ease!)
 - **Leading questions** that make assumptions, for example, why do you like ...?
 - **Unconscious biases**, for instance, gender stereotypes

Running the interview

- **Introduction:** Introduce yourself, explain the goals of the interview, reassure about the ethical issues, ask to record, and present the informed consent form.
- **Warm-up:** Make first questions easy and non-threatening.
- **Main body:** Present questions in a logical order
- **A cool-off period:** Include a few easy questions to defuse tension at the end
- **Closure:** Thank interviewee, signal the end, for example, switch recorder off.

Other forms of interviews

- Digital conferencing systems such as Skype, Zoom, email, and smartphones can be used to conduct interviews. Some advantages are:
 - Participants are in their **own environment** so are more relaxed
 - Participants **don't need to travel**
 - Participants **don't need to worry** about what to wear
 - For interviews involving sensitive issues, it is easier for interviewees to be **anonymous**

Asking better questions...



00:00 – 06:49 ALL

<https://www.growthdrivendesign.com/web-design/user-interviews>

Enriching the interview process

Props: Devices for prompting interviewee, for example, use a prototype, scenario



Let's examine this some more ...

When displaying text, what foreground/background colour combination is the easiest for you to read?

Background colour	Foreground Colour															
Yellow	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
Dark red	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
red	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
purple	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
pink	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
orange	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
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brown	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
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Black	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow

02:00

Question 1a: Colour choices

When displaying text, what foreground/background colour combination is the easiest for you to read?

Question 1b: Colour choices

When displaying text, what foreground/background colour combination is the easiest for you to read?

Background colour	Foreground Colour															
Yellow	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	
Dark red	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red		Yellow
red	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red		Yellow
purple	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
pink	Black	White	Blue	Dark blue	brown	Cyan	Dark cyan	grey	Dark grey	green	orange	pink	purple	red	Dark red	Yellow
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Question 1a: Colour choices

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Questionnaires

- We have already said that questions can be **closed or open**
- Closed questions are **easier to analyze**, and may be distributed and analyzed by computer
- They can be **administered to large populations**
- Disseminated by paper when? email and the web
- **Sampling can be a problem** when the size of a population is unknown as is common in online evaluation

Questionnaire design

- The **impact** of a question can be **influenced** by **question order** **overcome this?**
- You may need **different versions** of the questionnaire for different populations.
- **Provide clear instructions** on how to complete the questionnaire.
- **Strike a balance** between using white space and keeping the questionnaire compact (this isn't easy).
- Avoid **very long** questions and questionnaires (participants get bored and may tick randomly just to get finished)
- **Decide** on whether phrases will all be positive, all negative, or mixed (and then **stick to that format** consistently).

Question and response format

- ‘**Yes**’ and ‘**No**’ checkboxes
- **Checkboxes** that offer many options
(supports multiple “checks”)
- Rating **scales**
 - **Likert** scales
 - asks how much **do you agree/disagree** with the statement
(usually 3, 5 or 7 options)

Sullivan, Gail & Artino, Anthony. (2013). Analyzing and Interpreting Data From Likert-Type Scales.
Journal of graduate medical education. 5. 541-2. 10.4300/JGME-5-4-18.

https://www.researchgate.net/publication/259879583_Analyzing_and_Interpreting_Data_From_Likert-Type_Scales

- **Semantic** scales
 - See rating image above right
 - Based on **connotations**. A **connotation** is an idea or a feeling that the word invokes.
- **Open-ended** responses
 - Why do these work so well on Menti?

Please rate this lecture:

Boring : : : : Interesting

Slow : : : : Fast

Easy to understand : : : : Complicated

Encouraging a good response



- Make sure that the **purpose of study is clear**
- Promise **anonymity**
- Ensure that questionnaire is well designed
- Offer a **short version** for those who do not have time to complete a long questionnaire
- If mailed, include a stamped, addressed envelope
- **Follow-up** with emails, phone calls, or letters
- Provide an **incentive**
- 25 percent response rate is good, 10-15% percent is often acceptable
 - So if you **want** to get **25 responses**, you need to send it to **200+ people**

Advantages of online questionnaires

- Relatively **easy** and **quick** to distribute
- Responses are usually received quickly
- No photocopying and postage **costs**
- Data can be **collected** in database for analysis
- Time required for data analysis is reduced
- Errors can be corrected easily

Example of an online questionnaire

World Summit on the Information Society - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Address <http://www.itu.int/wsis/stocktaking/scripts/q.asp> Go

D. Internationally-agreed development goals outlined in the Millennium Declaration :

Is this activity relevant to achieving the MDGs listed below? (see www.un.org/millenniumgoals/ and the targets for each goal) ☒ Yes ☐ No
If yes, please tick all goals that apply

- ☐ Eradicate poverty and hunger
- ☐ Achieve Universal Primary Education
- ☐ Promote gender equality & empower women
- ☒ Reduce child mortality
- ☐ Improve maternal health
- ☐ Combat HIV/AIDS, Malaria and other diseases
- ☐ Ensure environmental sustainability
- ☐ Develop a global partnership for development

E. More Information :

Please provide a website for this activity
Website (URL) :

F. Geographical Coverage* :

Please tick a box to indicate the geographical coverage
☐ Local ☐ National ☐ Regional ☒ International
Please specify coverage :

G. Timescale* :

Please tick a box to indicate the timescale of the activity
☐ Completed ☐ Planned for future ☐ Ongoing

Specify dates using the format day/month/year (dd/mm/yyyy) :
From: To:

H. Activity Type* :

Please tick one or more boxes to indicate the type of activity described above
☐ Project ☐ Programme ☐ WSIS Thematic Meeting ☐ Conference ☐ Publication ☐ Training initiative
☐ Guidelines ☐ Tool-kit ☐ Website ☐ Database
Other (please specify) :

Questionnaire shows check boxes, radio boxes, and pull-down menus

Problems with online questionnaires

- Sampling is problematic if population size is unknown
- Preventing individuals from **responding more than once** can be a problem
- Individuals have also been known to change questions in email questionnaires
- Answer the next question (honestly) ...?

Deploying online questionnaires

- **Plan** the timeline
- Design offline
- Program/complete online template
- **Test the survey** to make sure that it behaves as you would expect
- **Test it with a group** that will not be part of the survey to check that the questions are clear
- **Recruit participants**

Observation

- **Direct observation** in the field
 - Structuring frameworks
 - Degree of (insider or outsider)
 - Ethnographic participation (culture based)
- **Direct observation** in **controlled** environments
- **Indirect observation**: tracking users' activities
 - Diaries
 - Interaction **logging**
 - Video and photographs **collected remotely** by drones or other equipment

Observation



Figure 8.8
Mars Exploration Rover

Source: Reproduced by permission of NASA Jet Propulsion Laboratory (NASA-JPL).

Main problem identified was scientists limitations in displaying, sharing and storing information

Structuring frameworks to guide observation

- Three easy-to-remember parts:
 - The person: **Who**? The place: **Where**? The thing: **What**?
- A more detailed framework presented by Robson:
 - **Space**: What is the physical space like and how is it laid out?
 - **Actors**: What are the names and relevant details of the people involved?
 - **Activities**: What are the actors doing and why?
 - **Objects**: What physical objects are present, such as furniture
 - **Acts**: What are **specific individual actions**?
 - **Events**: Is what you observe part of a special event?
 - **Time**: What is the **sequence** of events?
 - **Goals**: What are the actors trying to accomplish?
 - **Feelings**: What is the **mood** of the group and of individuals?

Planning and conducting observation in the field

- Decide on how involved you will be: from **passive observer** to **active participant**
- How to **gain acceptance**
- How to **handle sensitive topics, for example, culture, private spaces**, and so on
- How to collect the data:
 - What **data** to collect
 - What **equipment** to use
 - **When** to stop observing

Observations and materials that **might** be collected

Crabtree, A. (2003) *Designing Collaborative Systems: A Practical Guide to Ethnography*. Springer-Verlag, Berlin.

- Activity or **job descriptions**
- Rules and procedures that govern particular activities
- Descriptions of activities observed
- Recordings of the talk taking place between parties
- **Informal interviews** with participants explaining the detail of observed activities
- **Diagrams** of the **physical layout**, including the position of artifacts
- Other information collected when observing activities:
 - Photographs of artifacts (documents, diagrams, forms, computers, and so forth)
 - Videos of artifacts
 - Descriptions of artifacts
 - Workflow diagrams showing the sequential order of tasks
 - Process maps showing connections between activities

Direct observation in a controlled environment

- **Direct** observation
 - Think aloud techniques mention PhD examination @ UEA on this topic = a whole PhD!
 - Continuous Thinking aloud – whole audio is automatically transcribed and classified – interesting!

J. O. Johanssen, L. M. Reimer and B. Bruegge, "Continuous Thinking Aloud," 2019 IEEE/ACM Joint 4th International Workshop on Rapid Continuous Software Engineering and 1st International Workshop on Data-Driven Decisions, Experimentation and Evolution (RCoSE/DDrEE), Montreal, QC, Canada, 2019, pp. 12-15, doi: 10.1109/RCoSE/DDrEE.2019.00010.

- **Indirect** observation – tracking users' activities
 - Diaries
 - Interaction logs
 - Web analytics
- Video, audio, photos, and notes are used to capture data in both types of observations



Activity

Observation Questions

1. Are there cars parked on the sides of the road?
2. What colour is the pickup truck driving in the road?
3. What does the blue sign say?
4. What's the speed limit?
5. Are there any pedestrians on the road?

The Big One - Choosing and combining techniques

- Depends on the:
 - Focus of the study
 - Participants involved
 - Nature of the technique(s)
 - Resources available
 - Time available
- So often it comes down to **what can you afford** to do?

**What should I do
with the data
gathered?**

Quantitative and qualitative data

- **Quantitative** data: Expressed as numbers
- **Qualitative** data: Difficult to measure sensibly as numbers, for example, count number of words to measure dissatisfaction
- **Quantitative analysis**: Numerical methods to ascertain size, magnitude, and amount
- **Qualitative analysis**: Expresses the nature of elements and is represented as themes, patterns, or stories
 - Marco's work?
- **Be careful** how you manipulate data and numbers!

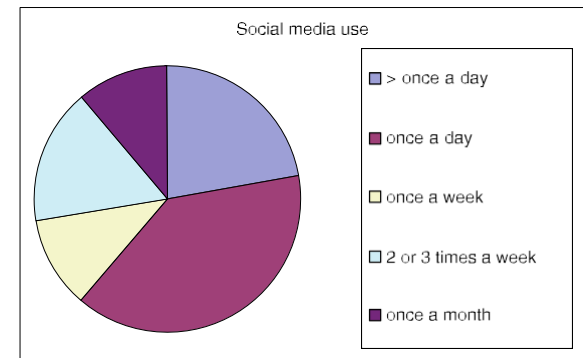
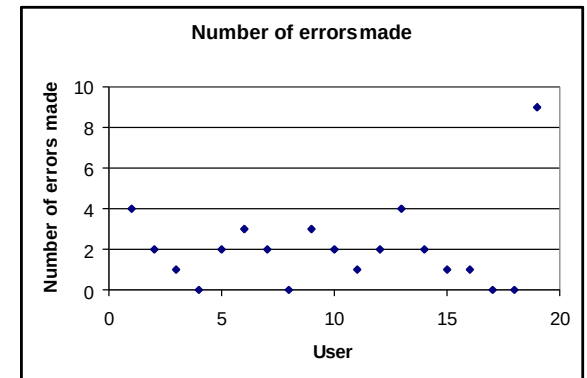
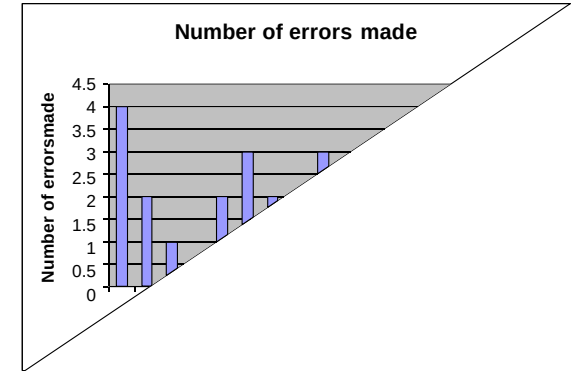
Basic quantitative analysis

- **Averages**

- **Mean**: Add up values and divide by number of data points
- **Median**: Middle value of data when ranked
- **Mode**: Figure that appears most often in the data

- **Percentages**

- Be **careful not to mislead** with numbers!
- Graphical representations give overview of data (**visualizations** are our universal language 😊)



How **question design** affects data analysis

- Question design affects analysis
 - Open question: Each answer analyzed separately
 - Closed question: Analyzed quantitatively
- **Fixed** alternative answers **restrict** what can be said in findings

Basic qualitative analysis

- Looking for **critical incidents**
 - Helps to focus in on **key** events
 - Then analysis can proceed using specific techniques
- Identifying **themes**
 - Emergent from data, dependent on observation framework if used
- Inductive analysis
- **Categorizing** data
 - Categorization scheme pre-specified
 - Deductive analysis
- In practice, combination of inductive and deductive

Interaction Analysis

- One way to **investigate** and **understand** interactions between people and between people and artefacts
- Based on empirical **observations** such as videos
- **Inductive** process in teams, collaboratively
- Contents of the material is logged
- Materials are extracted, classified, or removed
- Instances of a **salient** event are **assembled** and played one after the other
- The team of researchers studies the assemblage together

Tools to support data analysis

- **Spreadsheet** — Simple to use, basic graphs
- Statistical packages, for example, SAS and **SPSS**
- Qualitative data analysis tools
 - Categorization and theme-based analysis
 - Quantitative analysis of text-based data
- Nvivo and Dedoose support qualitative data analysis
- [Computer Assisted Qualitative Data Analysis \(CAQDAS\) Networking Project, based at the University of Surrey](#)

Interpreting and presenting the findings

Smartphone use
visualized as a
'bubble' diagram
(uses size encoding)

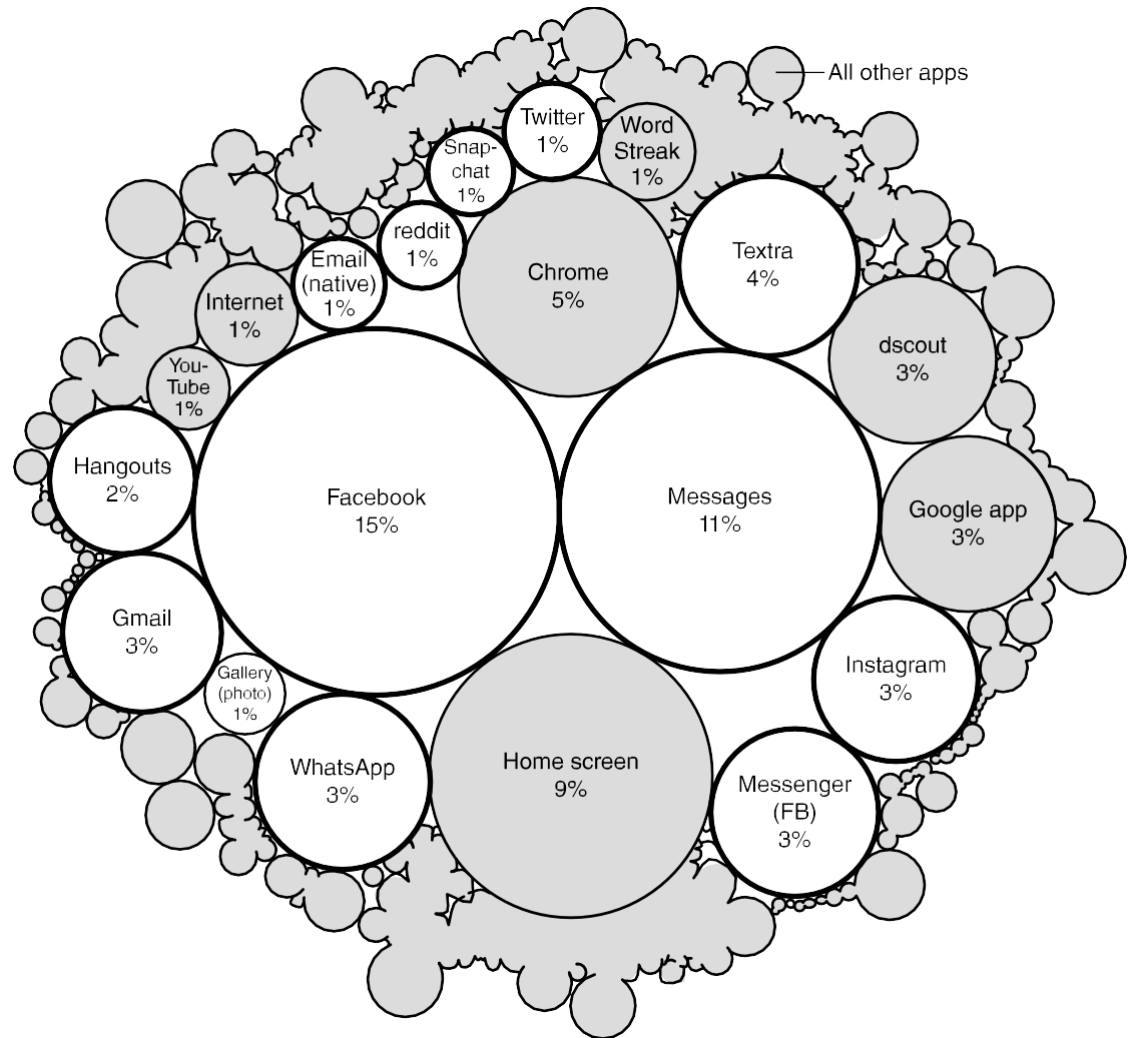


Figure 9.14 from the course textbook

Interpreting and presenting the findings

- Smartphone use visualized using a timeline of 24 hours
- It shows smartphone use for an average user vs a heavy user

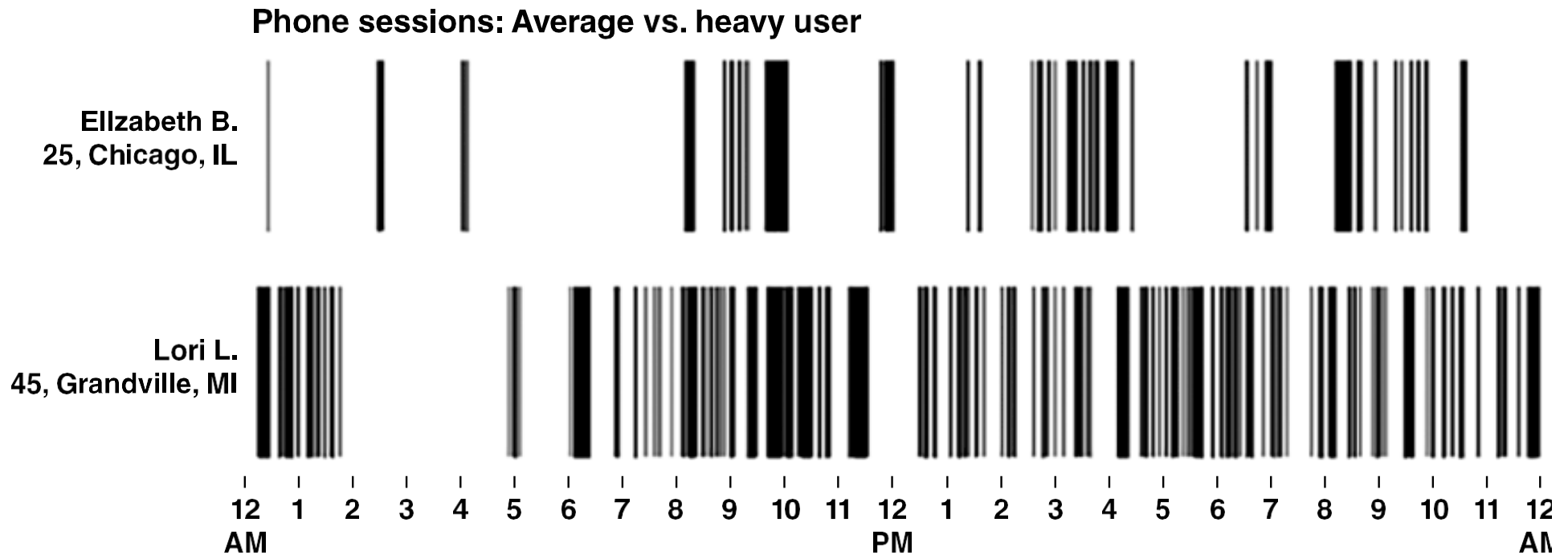


Figure 9.14 from the course textbook

Presenting findings

- **Structured notations** have clear syntax and semantics to present particular viewpoint
- **Stories** are easy and intuitive approach to communicate ideas
- **Summarize** findings using a **range of notations**



Later in this module we will study **data visualization** in significantly greater depth

Thank you